WEATHER REPORT: CONTEXT-AWARE PLAYLIST GENERATION FROM PHOTOS

Jackson MacNeil
University of Victoria
jacksonmacneil@uvic.ca

Brooklynn Nelson
University of Victoria
brooklynnn@uvic.ca

Jaiya Chauhan
University of Victoria
jaiyachauhan@uvic.ca

ABSTRACT

We present the current state of *Weather Report*, a system that turns a photo of the listener's surroundings and a target duration into a playlist whose mood matches the scene. Photos and songs are mapped into a shared, interpretable three-axis space: *warmth*, *happiness* (valence) and *energy* (arousal). On the image side, a zero-shot CLIP scorer rates each axis with a pair of opposing text prompts and routes the warmth judgement through an indoor/outdoor classifier. On the music side, we extract tempo, loudness, onset rate, spectral and key/mode features with librosa, compute CLAP zero-shot scores, and are building a hand-labelled seed set drawn from MTG-Jamendo to train a lightweight model. This report describes the revised design, the working components, preliminary results from prompt iteration and tempo validation, the status of each member's performance indicators, and our goals for the final project.

1. INTRODUCTION

Choosing music for a walk usually means queueing songs by hand. Streaming services recommend from listening history, but not from where the listener is or what the day feels like. *Weather Report* takes the opposite approach: the user uploads a photo of their environment and a desired playlist length, and the system returns a playlist whose mood and energy match the scene and, where available, the actual weather.

The core design decision is that **images and songs are described on the same scale**. Since our design specification we have reduced the planned six dimensions to three axes, each scored from 0 to 1: *warmth* (cold, grey and stark vs. sunny, cozy and inviting), *happiness* (sad vs. joyful) and *energy* (calm vs. lively). Happiness and energy correspond to the valence and arousal axes of Russell's circumplex model [1], which underpins most music emotion research [2, 3]; warmth is our own addition, motivated by the weather and context of the scene. Playlist generation then becomes a nearest-neighbour search in this space under a duration constraint.

We also revised two scope decisions. First, we do not train an LLark-scale model [4]; instead, "our own model" is a small regressor on top of librosa features and pretrained CLAP embeddings [5, 6]. Second, Spotify no longer provides audio features, recommendations or preview clips to new developer applications, so we analyse audio we can legally obtain (personal files, FMA [7] and MTG-Jamendo [8]) and treat streaming export as a stretch goal.

2. RELATED WORK

Music emotion recognition. Most music emotion recognition (MER) work models emotion in the valence–arousal plane [1]. Surveys link perceived emotion to audio features such as tempo, mode, loudness and timbre [2, 3], and DEAM provides continuous valence/arousal annotations for benchmarking [9]. Our feature extractor follows classic content-based MIR [10] using librosa [11], with beat tracking based on dynamic programming [12]. Tempo estimation is known to suffer from metrical-level (octave) errors [13], which we observe in our own validation (Section 5). Key and mode are estimated by correlating chroma with the Krumhansl–Kessler key profiles [14], and harmonic/percussive separation by median filtering [15] provides a drum-energy feature.

Joint embeddings of media and text. CLIP [16] learns a shared image–text embedding space and enables zero-shot classification with natural-language prompts; prompt wording strongly affects accuracy, motivating prompt ensembling and prompt learning [17]. CLAP [5, 6] and MuLan [18] apply the same contrastive idea to audio and music. LLark [4] connects music encoders to a large language model; it is out of scope for us but inspired the project.

Image–music matching and context. The closest prior work maps images and music into valence–arousal space and matches them there, introducing the IMEMNet dataset [19]; MMVA extends this with CLIP image features, a music encoder and musical captions [20]. We follow the same shared-space idea but use interpretable zero-shot axes, add a warmth axis, and plan to incorporate environmental context (time, location and weather), a direction surveyed in contextual music recommendation [21]. Playlist generation and its evaluation are surveyed in [22].

3. SYSTEM OVERVIEW

Figure 1 shows the pipeline. Six components (C1–C6) were defined in the design specification and are owned two per person (Section 6).

All code is written in Python 3.11 using `open_clip`, `laion_clap`, `librosa`, `scikit-learn` and SQLite. Components communicate through a shared output format: a dictionary with keys *warmth*, *happy* and *energy*, each a

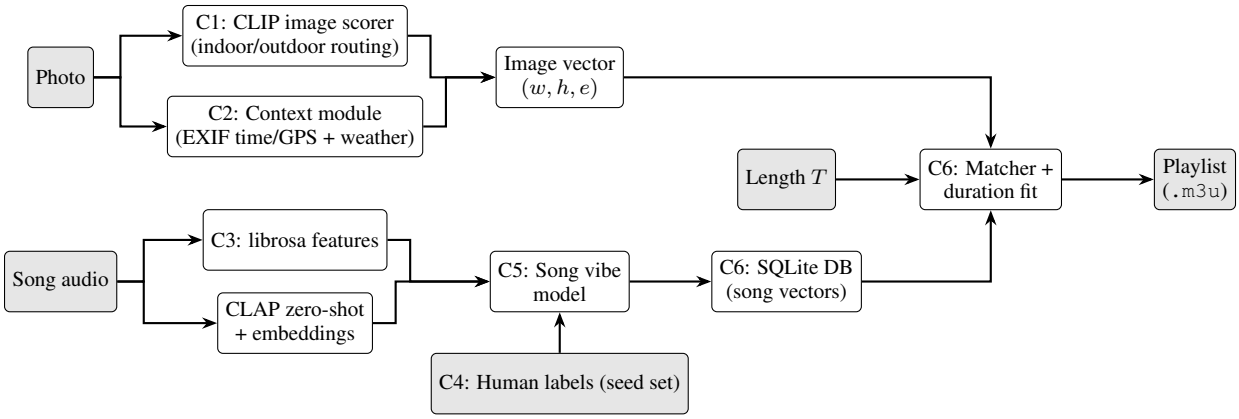


Figure 1. The *Weather Report* pipeline. Grey boxes are inputs/outputs. Photos and songs are both mapped to a warmth/happiness/energy vector and matched in that shared space.

float in $[0, 1]$. The image scorer (`image_to_vector`) and the CLAP scorer (`song_to_vector`) already return this format; the librosa extractor (`extract_features`) returns raw features that the song model maps into it. [TODO: Jackson: replace with the final interface list from D3.]

4. METHODS

4.1 Zero-Shot Image Scoring (C1)

We use CLIP (ViT-B/32, LAION-2B weights) zero-shot. The photo and each text prompt are embedded and compared by cosine similarity. Each axis is defined by a pair of opposing caption-style prompts, e.g. "a photo of a sad, lonely moment" vs. "a photo of a happy, joyful moment", and the axis score is the softmax probability of the positive prompt within that pair. Scoring each pair separately, rather than one softmax over all prompts, lets a photo be both happy *and* calm. A temperature (scale) factor controls how extreme the scores are: CLIP's default of 100 produced near-binary outputs, so we use 25 for the mood axes.

Warmth proved hardest (Section 5). Prompts about colour temperature mistook a blue-sky beach for "cold", while prompts about weather cannot judge indoor scenes. We therefore first classify the photo as indoor or outdoor (with a sharper scale of 50). Outdoor photos are scored with weather prompts ("cloudy, rainy day" vs. "clear, sunny day"); indoor photos with colour/coziness prompts ("cold and stark, with blue and grey colours" vs. "warm and cozy, with soft colours and light").

4.2 Audio Features (C3)

The extractor loads each file at 22.05 kHz mono and computes features grouped by the axis they are expected to inform: *energy* – tempo (beat tracking [12]), mean RMS loudness (dB) and onset rate (onsets/s); *warmth* – spectral centroid ("brightness"), share of power in the 100–800 Hz band and spectral flatness ("noisiness"); *happiness* – mode and mode strength, computed by correlating the song's average chroma vector with all twelve rotations of the Krumhansl–Kessler major and minor key profiles [14]; mode strength is the best major correlation minus the best minor correlation, so positive values lean major. [TODO: Jaiya: add any features added since (MFCCs, duration, HPSS ratio).]

4.3 CLAP Scoring and Song Model (C4, C5)

CLAP [6] scores songs zero-shot with the same pairwise prompt scoring as the image side, using audio prompts such as "a sad and melancholic song" vs. "a happy and cheerful song" (scale 20). This is our untrained baseline (System A). Systems B and C will regress human labels from librosa features alone and from librosa features plus the 512-dimensional CLAP embedding (reduced with PCA), using ridge regression, random forests and a small MLP. Labels are collected on MTG-Jamendo tracks, whose mood/theme tags serve as a sanity check. [TODO: Brooklynn: labelling protocol, scale (currently Low/Mid/High), number of songs labelled, rater agreement.]

4.4 Database and Playlist Assembly (C6)

A shared SQLite schema stores songs, audio features, CLAP embeddings, human ratings (with a consensus view), model scores per system and generated playlists. Given a photo's vector and a target length, the assembler ranks songs by Euclidean distance between the photo and song vectors and adds the closest songs until the total length is within 3 minutes of the target. [TODO: Jaiya: update once the assembler is implemented; note schema migration from six to three axes.]

5. PRELIMINARY RESULTS

5.1 Image Scorer

[TODO: Jackson: image scorer results table on the photo test set (D5), with a short discussion. Development-photo observations so far: the foggy forest scores low on all three axes, the party and pub score high on happiness and energy,

Warmth prompt pair	Beach	Party	Pub	Harb.	Fog
Colour (blue/grey vs. golden)	0.21	0.80	0.54	0.58	0.01
Stark vs. cozy/comforting	0.26	0.35	0.46	0.24	0.20
Dark/gloomy vs. bright/clear	0.92	0.92	0.81	0.87	0.12
Mean of colour + feeling	0.34	0.71	0.49	0.50	0.10
Cloudy/rainy vs. clear/sunny	0.93	0.78	0.50	0.65	0.08
Indoor/outdoor routed (final)	0.93	0.78	0.64	0.65	0.08

Table 1. Warmth scores for different prompt designs.

Feature	*Sex on Fire*	*Fake Plastic Trees*
Tempo (BPM)	152.0	73.8
Loudness (dB, mean RMS)	-15.1	-16.5
Onset rate (1/s)	3.25	2.36
Spectral centroid (Hz)	2263	1625
100–800 Hz power share	0.41	0.73
Spectral flatness	0.018	0.010
Mode / strength m	major / 0.03	major / 0.28

Table 2. librosa features for two example songs.

the overcast harbour lands near the middle, and the night-time backyard party is the one borderline indoor/outdoor case.]

Prompt iteration for warmth. Table 1 summarises how the warmth axis evolved. Colour prompts called the sunny beach cold (0.21); "cozy" prompts favoured indoor scenes and pushed every outdoor photo cold; light-only prompts ("dark and gloomy" vs. "bright and clear") rated every photo except the fog above 0.8. Averaging the scores of two prompt pairs widened the spread, and weather prompts plus indoor/outdoor routing gave the ordering closest to our judgement. Earlier, single-word prompts ("warm", "sad") and prompts mixing axes ("happy, chill or relaxed") produced confounded scores.

5.2 Audio Features and Tempo Validation

Table 2 shows features for two contrasting songs. The rock track is faster, has more onsets per second and is brighter; the ballad concentrates more power in the 100–800 Hz "body" band and has a stronger major-mode score. Tempo, however, is unreliable on slow, syncopated songs: for *Superpowers* (Daniel Caesar), librosa estimated 107.7 BPM and Tunebat (also algorithmic) reported 107 BPM, whereas a hand-tapped estimate and the Hooktheory transcription both give 80 BPM. The ratio of about 4/3 is a metrical-level error that a simple octave fold cannot correct, so tempo will be one of several energy cues rather than the main one. [TODO: Jaiya: full 20-song tempo validation table (D5).]

[TODO: Brooklynn: CLAP zero-shot results on the labelled MTG-Jamendo songs (e.g., agreement between CLAP Low/Mid/High and our labels).]

6. INDIVIDUAL STATUS REPORTS

Status: Done = complete; Partial = started, not complete; Not started = not yet started.

ID	Tier	Performance indicator	Status
D1	Basic	Run CLIP locally; photo–prompt similarity	Done
D2	Basic	Prompt sets per axis; 0–1 vector per photo	Done
D3	Basic	Shared interfaces between components	Partial
D4	Basic	Stub pipeline end to end on fake data	Not started
D5	Expected	30–50 photo test set; correlation with team ratings	Partial
D6	Expected	Weather/time context (EXIF + Open-Meteo)	Partial
D7	Expected	Single command: photo + length → playlist	Not started
D8	Advanced	Compare prompt strategies and CLIP model sizes	Partial
D9	Advanced	Listening study, 8–10 participants	Not started

Table 3. Jackson MacNeil – performance indicators.

6.1 Jackson MacNeil

D1 and D2 are complete: `Image/image_scorer.py` exposes `image_to_vector(path)`, which returns the three-axis vector using the pairwise scoring and indoor/outdoor routing of Section 4.1. The design moved from six dimensions to three after single-softmax experiments showed that several of the original dimensions were opposite ends of the same axis (e.g., energy and calmness). D5 is under way with six development photos; the indoor/outdoor classifier (part of D6) is working, while EXIF and weather lookup are next. D8 is partly covered by the prompt comparison in Table 1. Remaining work: formalise the interfaces (D3), build the stub pipeline (D4), finish the photo test set and correlation (D5), add EXIF/weather context (D6), and connect everything into one command (D7). [TODO: Jackson: update statuses and wording before submission.]

6.2 Brooklynn Nelson

ID	Tier	Performance indicator	Status
D1	Basic	Labelling protocol with rating scale and example songs	[TODO:]
D2	Basic	Labelling tool that plays a clip and saves ratings	[TODO:]
D3	Basic	CLAP zero-shot scores and embeddings for every song	[TODO:]
D5	Expected	∼100-song seed set; coordinate labelling by all 3	[TODO:]
D6	Expected	Model ratings vs. group ratings	[TODO:]
D7	Expected	Compare Systems A, B and C	[TODO:]
D8	Expected	Compare ridge, random forest, MLP; feature importance	[TODO:]
D9	Advanced	Train on a large dataset; compare with LLark	[TODO:]

Table 4. Brooklynn Nelson – performance indicators.

[TODO: Brooklynn: describe progress in your own words. Current repo state: `Song/song_scorer.py` computes CLAP zero-shot scores on the three axes (scale 20, adjusted energy prompt); `Song/labeled_test_songs.csv` holds the first eight MTG-Jamendo tracks labelled Low/Mid/High on

energy, happiness and warmth. Add what is done, what is left, and what you learned.]

6.3 Jaiya Chauhan

ID	Tier	Performance indicator	Status
D1	Basic	librosa: tempo, loudness, key/mode, spectral, MFCCs, duration	[TODO:]
D2	Basic	SQLite schema; store features for the seed set	[TODO:]
D3	Basic	Rank songs by similarity to the photo vector	[TODO:]
D4	Basic	Fit total length within ±3 min; `.m3u` export	[TODO:]
D5	Expected	Validate tempo vs. hand-tapped BPM on 20 songs	[TODO:]
D6	Expected	Process a 300+ song library	[TODO:]
D7	Expected	UI for photo upload, length and playlist display	[TODO:]
D8	Expected	Playlist flow (no repeat artists, smooth energy)	[TODO:]
D9	Advanced	Deploy a web app; streaming export if possible	[TODO:]

Table 5. Jaiya Chauhan – performance indicators.

[TODO: Jaiya: describe progress in your own words. Current repo state: `Librosa/extract_features.py` extracts tempo, loudness, onset rate, spectral centroid, low-mid power share, spectral flatness and mode/mode strength into `features.csv`; `Database/schema.sql` and `db.py` define the shared SQLite database (still using the original six dimensions). Add what is done, what is left, and what you learned.]

7. REVISED TIMELINE AND FINAL GOALS

The original timeline assumed six dimensions and a single labelling pass; both changed. Table 6 gives the revised plan. [TODO: All: confirm dates against the course deadline.]

Basic (deliverable even with delays): a command-line tool that takes a photo and returns a ranked list of songs from a library of at least 100 tracks, using the CLIP image scorer and CLAP zero-shot song scores.

Expected: the full pipeline in one command (`python weather_report.py photo.jpg –minutes 45`) with weather/time context, a trained song model, a 300+ song library, duration-matched `.m3u` output and a simple UI; evaluation of Systems A/B/C against held-out human labels and of the image scorer against team ratings of a photo test set.

Advanced: a blind listening test comparing our playlists with zero-shot and random baselines; a hosted web version; direct playlist export to a streaming service if API access allows.

8. LESSONS LEARNED

Prompt design is the model design. Single words, mixed-axis vocabulary and unbalanced prompt pairs all produced misleading scores; describing what a scene *looks like* worked better than naming an abstract feeling. *Every judge has blind spots.* Colour, light and weather prompts each failed on different photos, and combining them (by averaging or routing) helped more than tuning any single prompt. *Beware of tuning to the test set.* Moving the indoor/outdoor threshold to fix one photo would not generalise; a larger photo test set rated by the team is needed. *Algorithms can agree and both be wrong,* as the tempo example shows; human ground truth is essential. Finally, external APIs change: the Spotify restrictions forced us to own the whole audio-analysis pipeline.

Week	Milestone
Nov 16	Shared interfaces + stub pipeline; DB migrated to 3 axes; 100 seed songs labelled by all three members
Nov 23	End-to-end CLI; train Systems B/C; photo test set rated; EXIF + weather context
Nov 30	A/B/C comparison; playlist assembler on 300+ songs; simple UI; `.m3u` export
Final	Evaluation, listening test (if time), final paper

Table 6. Revised timeline.

9. AI USE DISCLOSURE

We used Claude (Anthropic), an AI assistant, in preparing this report. It produced the TikZ code for the pipeline diagram (Figure 1), set up the ISMIR LaTeX template and helped with formatting the report. It was also used as a tutor and coding assistant while developing the image scorer. Finally it helped draft MD files to save time so the team can focus on coding, helped organise our git branches so we did not lose our progress. All system design decisions, experiments, and reported results are the authors' own. [TODO: Brooklynn, Jaiya: add any AI tools you used for your components.]

10. REFERENCES

[1] J. A. Russell, "A circumplex model of affect," *Journal of Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.

[2] Y. E. Kim, E. M. Schmidt, R. Migneco, B. G. Morton, P. Richardson, J. Scott, J. A. Speck, and D. Turnbull, "Music emotion recognition: A state of the art review," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2010.

[3] R. Panda, R. Malheiro, and R. P. Paiva, "Audio features for music emotion recognition: A survey," *IEEE Trans. Affective Computing*, vol. 14, no. 1, 2023.

[4] J. Gardner, S. Durand, D. Stoller, and R. M. Bittner, "LLark: A multimodal instruction-following language model for music," in *Proc. Int. Conf. Machine Learning (ICML)*, 2024.

[5] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, "CLAP: Learning audio concepts from natural language supervision," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

[6] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

[7] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, "FMA: A dataset for music analysis," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2017.

[8] D. Bogdanov, M. Won, P. Tovstogan, A. Porter, and X. Serra, "The MTG-Jamendo dataset for automatic music tagging," in *Proc. ICML Workshop on Machine Learning for Music Discovery*, 2019.

[9] A. Aljanaki, Y.-H. Yang, and M. Soleymani, "Developing a benchmark for emotional analysis of music," *PLOS ONE*, vol. 12, no. 3, 2017.

[10] G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," *IEEE Trans. Speech and Audio Processing*, vol. 10, no. 5, 2002.

[11] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in Python," in *Proc. 14th Python in Science Conf. (SciPy)*, 2015, pp. 18–25.

[12] D. P. W. Ellis, "Beat tracking by dynamic programming," *Journal of New Music Research*, vol. 36, no. 1, 2007.

[13] F. Gouyon, A. Klapuri, S. Dixon, M. Alonso, G. Tzanetakis, C. Uhle, and P. Cano, "An experimental comparison of audio tempo induction algorithms," *IEEE Trans. Audio, Speech, and Language Processing*, vol. 14, no. 5, 2006.

[14] C. L. Krumhansl, *Cognitive Foundations of Musical Pitch*. New York, NY, USA: Oxford University Press, 1990.

[15] D. FitzGerald, "Harmonic/percussive separation using median filtering," in *Proc. Int. Conf. Digital Audio Effects (DAFx)*, 2010.

[16] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021.

[17] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *Int. Journal of Computer Vision*, vol. 130, 2022.

[18] Q. Huang, A. Jansen, J. Lee, R. Ganti, J. Y. Li, and D. P. W. Ellis, "MuLan: A joint embedding of music audio and natural language," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2022.

[19] S. Zhao, Y. Li, X. Yao, W. Nie, P. Xu, J. Yang, and K. Keutzer, "Emotion-based end-to-end matching between image and music in valence-arousal space," in *Proc. ACM Int. Conf. Multimedia*, 2020.

[20] S. Choi, K. W. Kim, and M. Kang, "MMVA: Multimodal matching based on valence and arousal across images, music, and musical captions," arXiv:2501.01094, 2025.

[21] M. Kaminskas and F. Ricci, "Contextual music information retrieval and recommendation: State of the art and challenges," *Computer Science Review*, vol. 6, no. 2–3, 2012.

[22] G. Bonnin and D. Jannach, "Automated generation of music playlists: Survey and experiments," *ACM Computing Surveys*, vol. 47, no. 2, 2014.